



MINING DEALER BEST PRACTICE

789 – 793 FINAL DRIVE LIFTING TOOL

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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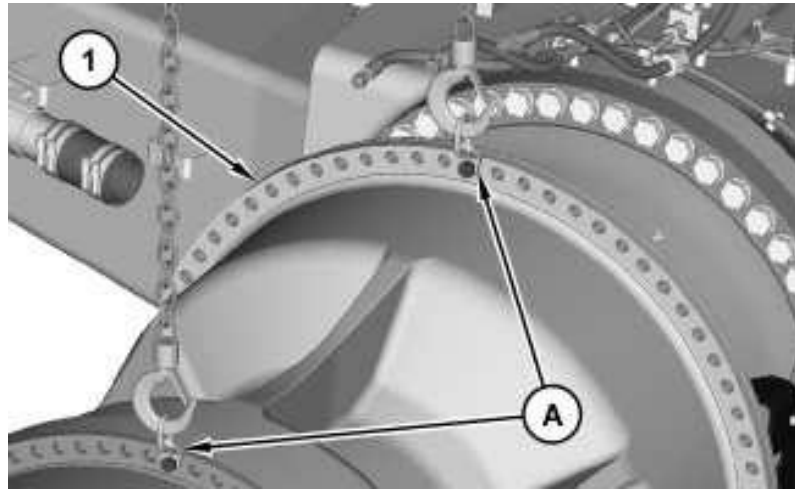
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1.0 Introduction

Part of the process of assembly and disassembly is based on handling components before and after passing through the CRC. This state makes for complex handling while in its original form, thus making it an important and critical task performed in workshops.

Based on the above premise arises the need to perform this task more efficiently. This best practice provides the technician a more efficient process by using a tool that provides security when performing the movement of the final drive.

This improvement is aims to prevent and provide security when moving the component.



This Best Practice aims to facilitate and expedite the movement of the 789 – 793D final drives by outlining a process for the manufacturing, use and implementation of an easy tool.

2.0 Best Practice Description

Each group consists of two triangular lifting plates – a rear (larger diameter) and a front (smaller diameter), both are ASTM A36 steel with a thickness of 1-inch. The lower arch molding with different diameters increases the stability of lifting the component. The main features of each are:

- Lifting Tool Final Drive 793-1
Dimension: 25.4mm x 248mm x 150mm (D x W x H)
Total weight: 4.26 Kg



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• Lifting Tool Final Drive 793-2

Dimension: 25.4mm x 223mm x 155mm (D x W x H)

Total weight: 4.05 Kg



• Lifting Tool Command Final 789-1

Dimension: 25.4mm x 223.5mm x 150mm (D x W x H)

Total weight: 3.79 Kg



• Lifting Tool Command Final 789-2

Dimension: 25.4mm x 200mm x 150mm (D x W x H)

Total weight: 3.56 Kg



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3.0 Implementation Steps

Step 1: Select appropriate tools the task. (Final Drive 789 or 793)

Step 2: Review the state of the tool to verify its integrity before use.

Step 3: After finishing the final drive assembly, place the lifting tool on the outer ring of the wheel, attaching it to the appropriate screws (see Figures 1 and 2).

Step 4: Using the Jib crane, lift the Final Drive to the desired height (see Figure 3).

Step 5: Place the component in its final housing (stowage basket, box) and remove the tools.



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4.0 Benefits

The main benefits provided by this implementation are:

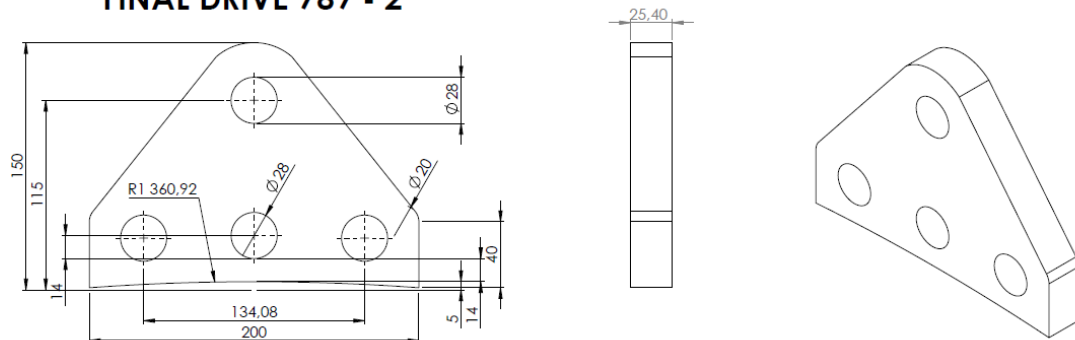
- Security when hoisting the final drive
- Reduced ergonomic risks for technicians
- Easy to perform the work

5.0 Resources Required

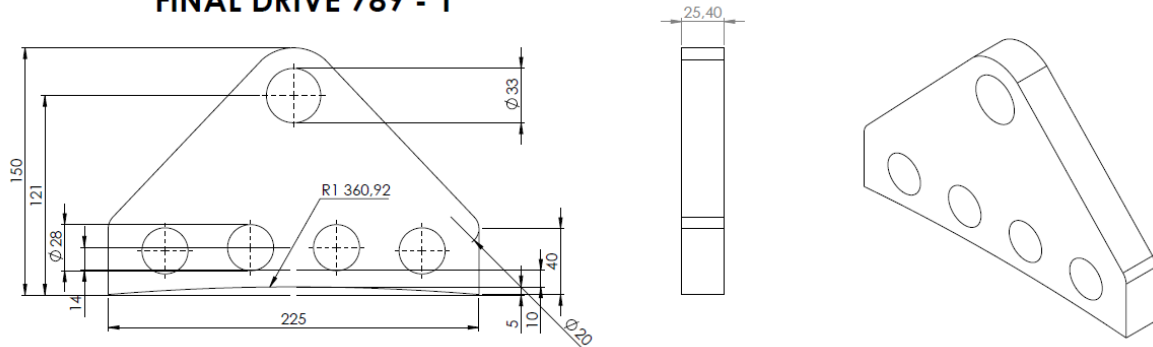
- Materials and manufacturing equipment according to plan
- 1 welder

6.0 Supporting Attachments / References

FINAL DRIVE 789 - 2



FINAL DRIVE 789 - 1



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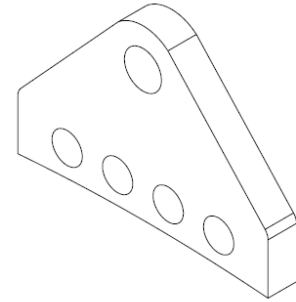
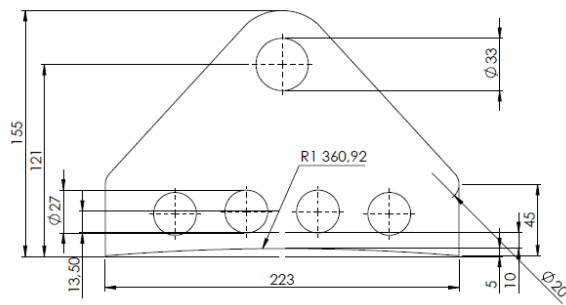
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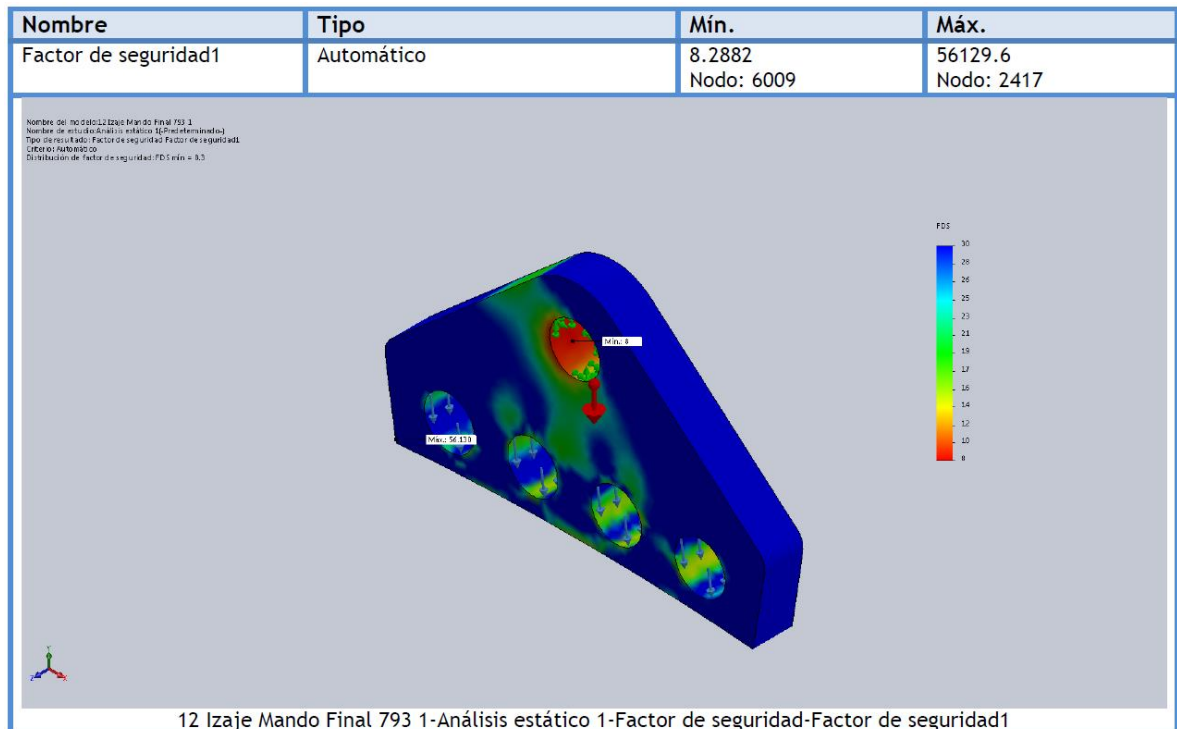
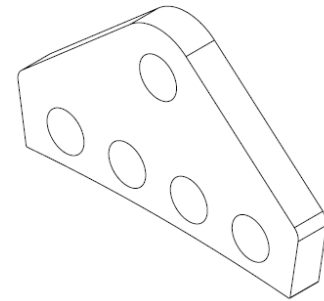
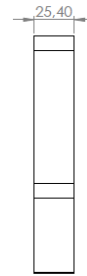
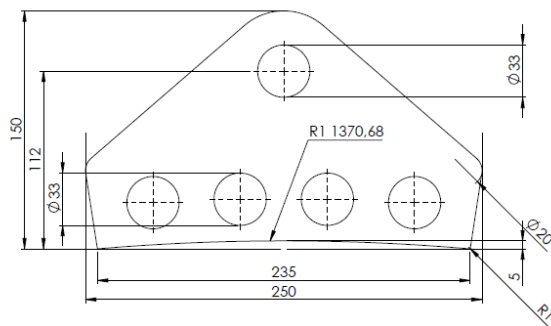
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FINAL DRIVE 793 - 2

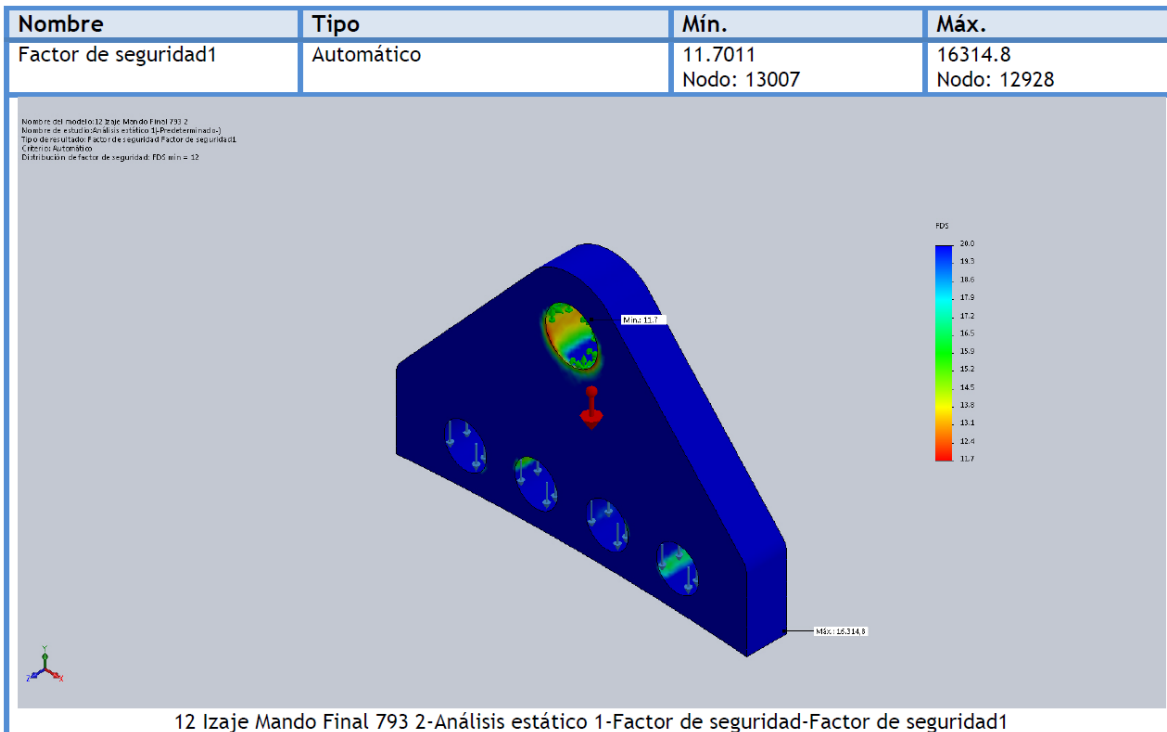


FINAL DRIVE 793 - 1

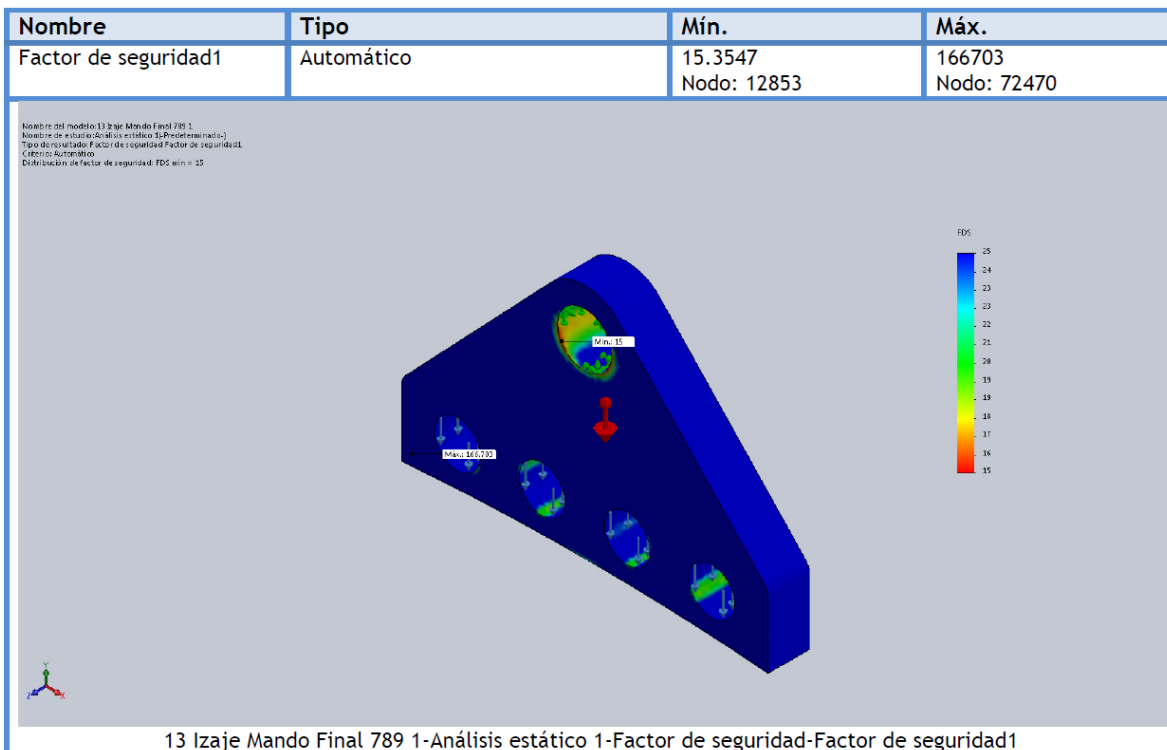


The minimum safety factor is 8, being greater than 5.0 FS, it is considered a safe tool for use. (Load 2419 Kg / 23719 N, applied to face contact holes tool holder with final drive.)

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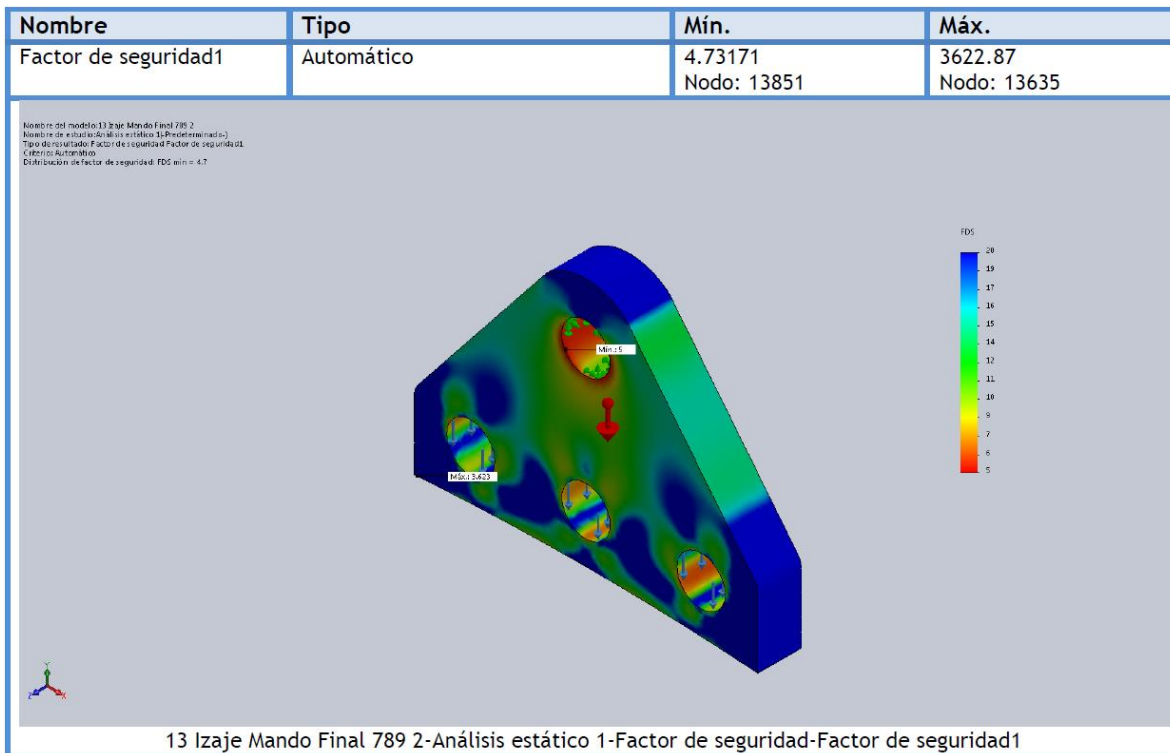


The minimum safety factor is 11.7, being greater than 5.0 FS, resulting in a safe tool for use. (Load 2419 Kg / 23719 N, applied to face contact holes tool holder with final drive.)



The minimum safety factor is 15.3, being greater than 5.0 FS, resulting in a safe tool for use. (Load 1778 Kg / 17433 N, applied to face contact holes tool holder with final drive.)

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The minimum safety factor is 5, as a FS equal to 5.0, resulting in a safe tool to use. (Load 1778 Kg / 17433 N, applied to face contact holes tool holder with final drive.)

7.0 Related Best Practices

None

8.0 Acknowledgements

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